

Electron Orbits from the Signed Electrostatic Force: Covalent, Metallic, and Intermolecular Magnitudes

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Abstract

On the equal-scale locus the electromagnetic force of double spacetime theory is the Coulomb force divided by the interference factor γ_s . The sign of γ_s is the repulsion. An unlike pair attracts where $\gamma_s > 0$ and repels where $\gamma_s < 0$; a like pair does the opposite. No centrifugal term is added to cancel an attraction. A single electron–proton pair then has no force-free radius: the radial force is nonzero wherever γ_s is finite. Two protons and two electrons do. In the first repulsive shell they sit at the vertices of a square, protons on one diagonal and electrons on the other, and the electrostatic potential has a positive-definite Hessian there. Like charges attract along the diagonals and unlike charges repel along the sides. The diagonal is $\sqrt{2}$ times the side, the diagonal force is $4.3532 ke^2/\ell^2$, and the stiffness of the diagonal is $8.178 ke^2/\ell^3$. A nearest-neighbor bond that carries only the fraction $2/z$ of that pair has further points where every electrostatic component vanishes. For eight and for twelve neighbors those points are saddles, and the partial forces stand in the ratios 0.04323 and 0.1704 to the covalent diagonal. Two neutral pairs a few side lengths apart keep a force after the monopole cancels. Identifying the side with the Bohr radius makes the diagonal $\sqrt{2}a_0$, against the Born–Oppenheimer separation $1.4011a_0$ of H_2 .

1 Introduction

The electron-shell section of double spacetime theory writes an electrostatic force whose denominator is the scalar interference factor γ_s [1]. At large separation the factor tends to one and the Coulomb law is recovered. Closer in, γ_s passes through zero on a discrete set of radii and is negative between those nodes. A node is a divergence of the force. The radial potential on the outer side of a node is unbounded below.

A purely attractive Coulomb force does not vanish at any finite radius. The classical circular orbit cancels it by a centrifugal term $L^2/(mr^3)$, the kinematics of a conserved angular momentum. That term is not added here. The outward contribution is already in the signed force. Where γ_s is negative an unlike pair repels, and where γ_s is positive a like pair repels. An electron orbit, in this note, is a locus on which these electrostatic contributions cancel. It is not a balance between Coulomb attraction and a centrifugal barrier.

The note computes that locus for a two-center covalent arrangement, for a nearest-neighbor bond that carries only $2/z$ of an electron pair, and for the residual force between two neutral pairs. The equal-scale identification $\alpha = \beta = \ell/r$ is the reading already used for the nuclear and Coulomb layers. The length ℓ is not derived. Bohr’s n^2 rule and the fine-structure constant are not derived. Where a laboratory number enters, it is marked as an input.

2 The signed force

On one axis the interference factor of the dual rotor is

$$\gamma_s = \cosh \frac{\alpha}{2} \cos \frac{\beta}{2} - \sinh \frac{\alpha}{2} \sin \frac{\beta}{2}.$$

The equal-scale locus sets $\alpha = \beta = \ell/r$. In the dimensionless variable $x = \ell/(2r)$,

$$\gamma_s(x) = \cosh x \cos x - \sinh x \sin x, \quad \gamma'_s(x) = -2 \cosh x \sin x. \quad (1)$$

Thus $\gamma_s(0) = 1$, and on each interval $[n\pi, (n+1)\pi]$ there is one simple zero. The first zero x_1 is the unique root of

$$\tan x = \coth x \quad (2)$$

in $(\pi/4, \pi/2)$. Numerically $x_1 = 0.93755$, so the outermost node lies at

$$\frac{r_1}{\ell} = \frac{1}{2x_1} = 0.53330.$$

The second zero lies at $x_2 = 3.9274$, and the inner edge of the first repulsive shell is $r_2/\ell = 0.12731$. Ordered by decreasing radius, the outer well $r > r_1$ has $\gamma_s > 0$, the shell $r_2 < r < r_1$ has $\gamma_s < 0$, and the sign continues to alternate. The extremum inside the first shell is $\gamma_s(\pi) = -\cosh \pi$, at $r = \ell/(2\pi)$.

The force that increases the separation of two charges is

$$F_r = \frac{kq_1q_2}{\gamma_s(r)r^2}, \quad k = \frac{1}{4\pi\epsilon_0}. \quad (3)$$

Like charges repel where $\gamma_s > 0$ and attract where $\gamma_s < 0$. Unlike charges do the opposite. Both directions are the same formula. Nothing further is added to produce a repulsion.

The Taylor expansion at the origin follows from $\gamma_s''(0) = -2$ and $\gamma_s'''(0) = -4$,

$$\gamma_s(x) = 1 - x^2 - \frac{1}{6}x^4 + O(x^6). \quad (4)$$

Hence

$$\frac{1}{\gamma_s(r)r^2} = r^{-2} + \frac{\ell^2}{4}r^{-4} + \frac{7\ell^4}{96}r^{-6} + O(r^{-8}).$$

The correction to the Coulomb force begins at order $(\ell/r)^2$.

For $r > r_1$ the pair potential that vanishes at infinity is the integral of (3). The substitution $s = \ell/(2r)$ gives $ds/s^2 = -(2/\ell)dx$, and therefore

$$\int_r^\infty \frac{ds}{\gamma_s(s)s^2} = \frac{2}{\ell} \int_0^{\ell/(2r)} \frac{dx}{\gamma_s(x)}, \quad (5)$$

provided the path stays in the outer well. The integral diverges as the upper limit approaches x_1 . Inside a shell the same formula does not continue: the path from infinity would cross a pole. A local force balance inside the shell is still well defined, because (3) requires only $\gamma_s \neq 0$.

3 No centrifugal term

Consider one proton and one electron. The radial force is

$$F_r = -\frac{ke^2}{\gamma_s(r)r^2}.$$

Wherever γ_s is finite and nonzero, F_r is nonzero. In the outer well it is attractive at every finite radius. In the first repulsive shell it is repulsive at every finite radius. There is no radius at which the electrostatic force vanishes, and none is introduced by hand.

The centripetal rearrangement $v^2 = ke^2/(m\gamma_sr)$ of the main paper records what a conserved angular momentum would contribute in the outer well. It is a kinematic identity for a circular Newtonian orbit. It is not an extra force in (3), and it is not used below. The repulsive shell already reverses the electron–proton force. At its quietest point, $x = \pi$,

$$|F| = \frac{4\pi^2}{\cosh \pi} \frac{ke^2}{\ell^2} = 3.4057 \frac{ke^2}{\ell^2}, \quad (6)$$

directed outward. That is the built-in repulsion, evaluated where $|\gamma_s|$ in the first shell is largest and the force is therefore smallest. A one-center electron is not parked on a force-free circle. A force-free locus appears only when several charges are present and their signed contributions cancel.

4 The square and the covalent force

Place protons of charge $+e$ at $z = \pm R/2$ and electrons of charge $-e$ at cylindrical radius ρ , opposite each other in the midplane. The three separations are the proton–proton distance R , the electron–electron distance $r_{ee} = 2\rho$, and the four equal electron–proton distances

$$r_{ep} = \sqrt{(R/2)^2 + \rho^2}.$$

No angular momentum is assigned. Equilibrium means that the axial force on each proton and the cylindrical force on each electron both vanish.

The axial force on the proton at $+R/2$, counted positive when it increases R , and the cylindrical force on one electron, counted positive when it increases ρ , are

$$F_z = ke^2 \left(\frac{1}{\gamma_s(R) R^2} - \frac{R}{\gamma_s(r_{ep}) r_{ep}^3} \right), \quad (7)$$

$$F_\rho = ke^2 \left(\frac{1}{\gamma_s(r_{ee}) r_{ee}^2} - \frac{2\rho}{\gamma_s(r_{ep}) r_{ep}^3} \right). \quad (8)$$

The first term of F_ρ is the electron–electron force. In the outer well it is a repulsion and points outward: that outward piece is electrostatic, and it occupies the place a centrifugal term would have occupied. In the repulsive shell $\gamma_s(r_{ee}) < 0$, so the same term points inward, because like charges attract there.

The configuration in which the four charges are the vertices of a square, with the protons on one diagonal and the electrons on the other, is

$$\rho = \frac{R}{2}, \quad r_{ee} = R, \quad r_{ep} = \frac{R}{\sqrt{2}}. \quad (9)$$

On this locus F_z and F_ρ reduce to one expression. With $x = \ell/(2R)$,

$$F_z = F_\rho = \frac{ke^2}{R^2} \left(\frac{1}{\gamma_s(x)} - \frac{2\sqrt{2}}{\gamma_s(\sqrt{2}x)} \right). \quad (10)$$

Both vanish when

$$\gamma_s(\sqrt{2}x) = 2\sqrt{2}\gamma_s(x). \quad (11)$$

In the first repulsive shell, where $x_1 < x < \sqrt{2}x < x_2$, there is one root,

$$x = 1.55463, \quad \frac{R}{\ell} = 0.32162, \quad \frac{r_{ep}}{\ell} = 0.22742.$$

The two interference factors are $\gamma_s(x) = -2.2208$ on the diagonals and $\gamma_s(\sqrt{2}x) = -6.2812$ on the sides. Every separation lies in (r_2, r_1) . Like charges therefore attract along both diagonals, and unlike charges repel along the four sides. The repulsion that holds the electrons off the protons is the side force. It is not a centrifugal force.

The partial forces at this root are

$$|F_{\text{diag}}| = \frac{ke^2}{|\gamma_s(x)| R^2} = 4.3532 \frac{ke^2}{\ell^2}, \quad (12)$$

$$|F_{\text{side}}| = \frac{ke^2}{|\gamma_s(\sqrt{2}x)| r_{ep}^2} = 3.0782 \frac{ke^2}{\ell^2}. \quad (13)$$

The diagonal force is the covalent force scale: it is the attractive force along the proton–proton separation, cancelled by the axial part of the side repulsions.

The electrostatic potential, regarded as a function of R and ρ near this root, has Hessian eigenvalues $37.33 ke^2/\ell^3$ and $7.933 ke^2/\ell^3$, both positive. The root is a local minimum. Allowing ρ to follow R , the stiffness of the diagonal is

$$\kappa = 8.178 \frac{ke^2}{\ell^3}. \quad (14)$$

Two further roots of $F_z = F_\rho = 0$ lie in the same shell. Each has Hessian eigenvalues of opposite sign, so each is a saddle. The square is the stable covalent orbit. Because the root lies inside the shell, the outer-well integral (5) does not assign it a dissociation energy relative to separated atoms at infinity.

5 Coordination and the metallic force

A nearest-neighbor bond in a monovalent coordination shell of z neighbors is given the electron content $\nu = 2/z$. The dimer is $z = 1$, $\nu = 2$, the square of the previous section. The charge $-\nu e$ is split into two lumps of charge $-(\nu/2)e$ at opposite points of the midplane. The forces (7) and (8) become

$$F_z = ke^2 \left(\frac{1}{\gamma_s(R) R^2} - \frac{\nu R}{2 \gamma_s(r_{ep}) r_{ep}^3} \right), \quad (15)$$

$$F_\rho = ke^2 \left(\frac{(\nu/2)^2}{\gamma_s(r_{ee}) r_{ee}^2} - \frac{\nu \rho}{\gamma_s(r_{ep}) r_{ep}^3} \right). \quad (16)$$

Equilibrium is still $F_z = F_\rho = 0$, with no centrifugal term. The outward piece in the outer well is the first term of F_ρ : the lumps repel where $\gamma_s(r_{ee}) > 0$.

For $z = 8$ and $z = 12$ that outer-well system has one root each. All three separations lie outside r_1 , so the signs are the ordinary ones: like charges repel, unlike charges attract. The partial proton–proton force and its ratio to the covalent diagonal (12) are

$$z = 8 : \quad \frac{R}{\ell} = 2.3591, \quad F_{\parallel} = 0.18820 \frac{ke^2}{\ell^2}, \quad \frac{F_{\parallel}}{|F_{\text{diag}}|} = 0.04323, \quad (17)$$

$$z = 12 : \quad \frac{R}{\ell} = 1.2666, \quad F_{\parallel} = 0.74188 \frac{ke^2}{\ell^2}, \quad \frac{F_{\parallel}}{|F_{\text{diag}}|} = 0.1704. \quad (18)$$

At both roots the Hessian of the potential has eigenvalues of opposite sign. Each is a saddle, not a local minimum. The same is true of the further roots that sit inside the repulsive shell. Reducing the electron content from a full pair to $2/z$ leaves a point where the signed forces cancel, and it does not leave a stable orbit. A crystal in which each ion is surrounded symmetrically can still have a vanishing net force by symmetry. That lattice sum is not the calculation above. An empirical nearest-neighbor distance would fix $F_{\parallel}$ by the proton–proton term alone; it is an input, and it is not required for the ratios.

6 Neutral pairs and the intermolecular force

The stable covalent orbit fixes an electron–proton distance $a = R/\sqrt{2} = 0.22742 \ell$. A neutral pair is a proton with an electron displaced by that distance. It is not a force-free one-center orbit: Section 3 has none. It is the side of the square, used as the size of a neutral pair. Two such pairs whose centers are a distance D apart have four intermolecular forces, each of the form (3). If each electron were coincident with its proton, every pair would share one distance and the sum would vanish with the net charge. The monopole cancels. What remains is the displacement a , read with the signed factor γ_s .

Two collinear placements are compared. In the parallel placement both electrons are displaced by $+a$ along the line of centers. In the inward placement the electrons lie between the protons. Attraction is counted positive. F is the net force on the left-hand pair and F_C is the same sum at $\gamma_s = 1$, so that the difference is the torsional excess. In units of the covalent diagonal,

D/a	$F/ F_{\text{diag}} $	$F_C/ F_{\text{diag}} $	$F/ F_{\text{diag}} $	$F_C/ F_{\text{diag}} $
	parallel		inward	
4	+0.5937	+0.1160	+4.537	−0.4010
6	+0.03668	+0.02155	−0.1056	−0.04565
8	+0.008803	+0.006679	−0.01666	−0.01149

At four side lengths the inward electron–electron separation is $2a = 0.4548 \ell$, which lies inside r_1 . That pair has $\gamma_s < 0$, so the two electrons attract, and the net inward force is several times the covalent diagonal while the pure Coulomb sum is repulsive. The amplification is the sign change in (3), not an extra interaction. At six and eight side lengths every intermolecular separation is outside the node. Both columns fall. At eight side lengths the parallel residue is 0.88% of the covalent diagonal, and about a quarter of that residue is the torsional excess over F_C .

7 A numerical illustration

The ratio of diagonal to side is $\sqrt{2}$ by (9), with no choice of ℓ . One identification turns the side into the Bohr radius,

$$a = r_{ep} = a_0.$$

The diagonal is then

$$R = \sqrt{2} a_0 = 1.4142 a_0,$$

against the Born–Oppenheimer equilibrium separation $1.4011 a_0$ of H_2 [3]. The predicted separation is long by a factor 1.009. The identification $a = a_0$ is an input; the factor $\sqrt{2}$ is the square. With the CODATA values of a_0 and of the Hartree energy $E_h = ke^2/a_0$ [2], the same identification gives $\ell = 4.3971 a_0$ and

$$|F_{\text{diag}}| = 18.55 \text{ nN}, \quad |F_{\text{side}}| = 13.12 \text{ nN}, \quad \kappa = 150 \text{ N m}^{-1}.$$

The diagonal force is the attractive proton–proton force at the equilibrium. The side force is the repulsive electron–proton force. The stiffness is (14). No dissociation energy is quoted: the equilibrium lies in the repulsive shell, and (5) does not cross the node. The measured dissociation energy of ortho- H_2 from $N = 1$ is 4.463 eV [4]; it is an external number, and it is not a value of this potential.

8 Conclusions

The sign of γ_s is the repulsion in (3). A centrifugal term is not required, and it is not introduced. One electron and one proton have no force-free radius. Two of each have a stable one: the square in the first repulsive shell, with diagonal force $4.3532 ke^2/\ell^2$ and stiffness $8.178 ke^2/\ell^3$. The bonds that carry 2/8 and 2/12 of an electron pair have electrostatic roots and indefinite Hessians; their partial forces are 0.04323 and 0.1704 of the covalent diagonal. Neutral pairs separated by eight side lengths retain a parallel force 0.0088 of that diagonal, and at four side lengths an inward pair enters the repulsive shell and the force changes by the sign of γ_s .

The square is a local minimum. The node still prevents the outer-well integral from assigning a dissociation energy, and the Bohr spectrum is not implied. One identification, the side equal to a_0 , converts the diagonal into $\sqrt{2} a_0$.

References

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